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WeChat-based manual enhanced guidance in patients with *Helicobacter pylori* infection

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Abstract

Background Patients' adherence is important for successful *Helicobacter pylori* (*H. pylori*) eradication, and may be improved by WeChat, a popular social networking platform. This study aimed to evaluate patients' adherence, successful *H. pylori* eradication, and clinical remission under the WeChat-based enhanced guidance.

Methods Ninety-nine patients with *H. pylori* infection, who were treated by a 14-day standard bismuth-containing quadruple therapy (BQT) and received the WeChat-based enhanced guidance, were retrospectively included. Multivariate logistic regression analysis was performed to identify independent predictors of patients' adherence and *H. pylori* eradication. Odds ratios (ORs) were calculated.

Results The rates of good medication adherence, good re-examination adherence, successful *H. pylori* eradication, and clinical remission were 97.0%, 83.8%, 89.2%, and 89.5%, respectively. Patients with poor re-examination adherence had significantly lower proportions of individual dining (13.3% versus 50.6%, *P*=0.008) and no smoking during treatment (66.7% versus 92.8%, *P*=0.012) than those with good re-examination adherence. Both individual dining (OR=0.182, P=0.035) and no smoking during treatment (OR=0.210, *P*=0.031) were independently associated with good re-examination adherence. Patients with *H. pylori* eradication failure had significantly higher proportions of poor medication adherence (22.2% versus 1.4%, *P*=0.030) and living in rural area (44.4% versus 8.1%, *P*=0.009) than those with *H. pylori* eradication success. Both poor medication adherence (OR=20.366, *P*=0.031) and living in rural area (OR=8.944, P=0.010) were independently associated with *H. pylori* eradication failure.

Conclusion There is good medication adherence with satisfactory *H. pylori* eradication and clinical remission under the WeChat-based enhanced guidance. How to improve re-examination adherence should be further explored.

Keywords Helicobacter pylori, Adherence, WeChat, Enhanced guidance, Eradication, Clinical remission

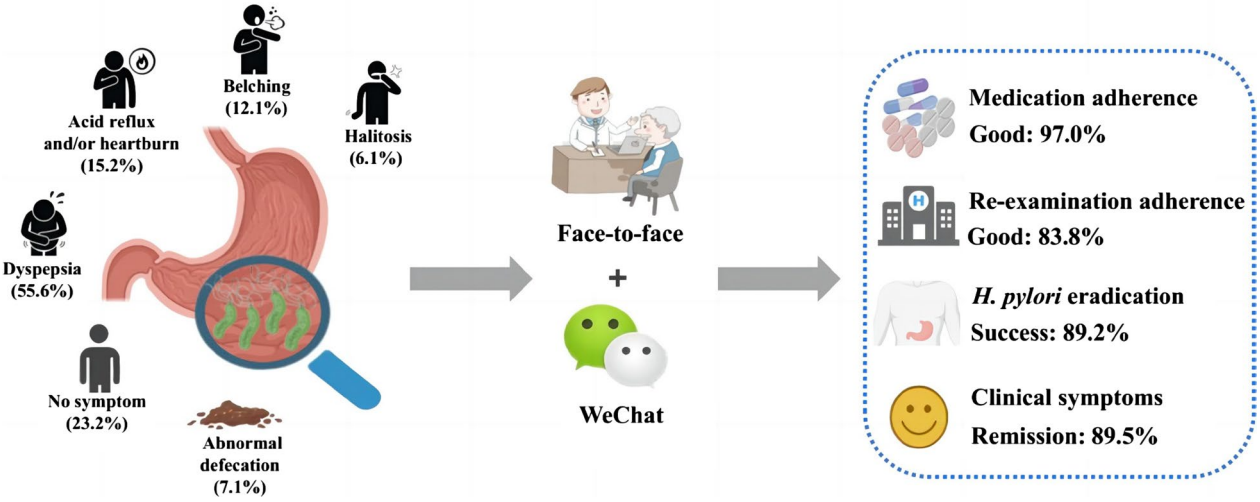
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Graphical Abstract



Introduction

Helicobacter pylori (*H. pylori*) infection is a well-recognized risk factor for chronic gastritis, peptic ulcer, mucosa-associated lymphoid tissue lymphoma, and gastric cancer [1]. Its prevalence ranges from 20.6% to 81.8% in the mainland region of China [2]. The Maastricht VI/Florence consensus has recommended that all *H. pylori* infected subjects, who are either symptomatic or asymptomatic, should receive *H. pylori* eradication therapy in principle [3]. Successful *H. pylori* eradication can effectively prevent from the development and progression of gastric lesions [4, 5]. Nowadays, a standard bismuth-containing quadruple therapy (BQT) regimen is strongly recommended as the preferred option for empirical *H. pylori* eradication in China [6], but it is still difficult to achieve an acceptable eradication rate of >85% in the real world [7, 8], which is probably related to poor medication adherence in some patients [9]. Considering that good adherence is a prerequisite for optimal outcome of any disease [10], there is an urgent need to improve the patients' adherence for *H. pylori* eradication.

Enhanced patient education is a convenient and cost-effective measure for improving patients' adherence [11]. Telephone-based re-education is a choice of enhanced education, but its benefit and acceptability are still controversial during *H. pylori* eradication therapy [12, 13]. With rapid development and popularization of the Internet, social medias have become indispensable for people's daily work and life, which fundamentally change the communications between people [14]. Nowadays, WeChat is one of the most popular social tool in China, with approximately 1 billion active users every day [15]. It not only combines the functions of phone call and text message, but also realizes remote medical care services by video between physicians and patients, especially during the coronavirus disease 2019 pandemic. The importance of WeChat on the management of chronic diseases has been gradually recognized [16].

Herein, we conducted this study to evaluate the patients' adherence, successful *H. pylori* eradication, and clinical remission under the WeChat-based enhanced guidance for *H. pylori* therapy, and to explore the predictors associated with patients' poor adherence and *H. pylori* eradication failure.

Methods

Study design

The study protocol has been approved by the Medical Ethical Committee of the General Hospital of Northern Theater Command with an approval number of Y (2023) 056 and follows the rule of the Declaration of Helsinki. The requirement for patients' written informed consents has been waived due to the retrospective nature of this study. We prospectively recorded and retrospectively analyzed the data of patients with *H. pylori* infection who were managed by two investigators (XQ and XS) at the Department of Gastroenterology of our hospital from November 2020 to September 2021. The inclusion criteria were as follows: (1) the age was not less than 18 years; (2) a 14-day BQT regimen was prescribed; (3) patients were willing to contact with an investigator (XS) via WeChat; and (4) patients verbally agreed to receive online enhanced guidance for *H. pylori* eradication. The exclusion criteria were as follows: (1) patients had taken medications for a duration of less or more than 14 days; (2) patients took probiotics during a 14-day BQT

regimen; and (3) patients' information on medication and/or re-examination could be not obtained. We have also selected patients receiving standard BQT regimens from our previous study [7], who were taken as the control group in the present study, to evaluate the potential efficacy of WeChat management on *H. pylori* therapy.

Diagnosis

In this study, *H. pylori* status was mainly identified by $^{13}C/^{14}C$-urea breath test ($^{13}C/^{14}C$-UBT) or stool antigen test. Among the treatment-naïve patients, if both of the two examinations were not applicable, serological test would be further performed. When any one of the three examinations was positive, *H. pylori* infection would be diagnosed.

Therapy

All patients were treated with the standard BQT regimen according to the fifth Chinese consensus on the management of *H. pylori* infection [6]. In detail, a clarithromycin-based therapy regimen (pantoprazole, 40 mg twice daily; colloidal bismuth pectin [CBP], 200 mg twice daily; clarithromycin, 500 mg twice daily, in combination with amoxicillin, 1000 mg twice daily [BPCA], or tetracycline, 500 mg three times daily [BPCT], or metronidazole, 400 mg three times daily [BPCM]) or a furazolidone-based therapy regimen (esomeprazole, 40 mg once daily; CBP, 200 mg twice daily; furazolidone, 100 mg twice daily; tetracycline, 500 mg three times daily [BEFT]) was prescribed according to the patients' history of *H. pylori* eradication and allergy to antibiotics, including amoxicillin, levofloxacin, and/or metronidazole. It is required that proton pump inhibitors (PPIs) and CBP were taken 30 minutes before meals, and antibiotics after meals. All of them should be administered orally for 14 days. Notably, in some patients with peptic ulcer and/or esophagitis, the duration of PPIs may be extended to 4–8 weeks or even longer. All patients were required to repeat $^{13}C/^{14}C$-UBT at least 4 weeks after discontinuation of PPIs and antibiotics.

Manual enhanced guidance

All patients were guided via both face-to-face and WeChat (Fig. 1).

Two physicians (XQ and XS) gave pre-medication educations via face-to-face before a 14-day BQT regimen. In detail, the harms of *H. pylori* infection and the necessity of *H. pylori* eradication were clearly explained. Meanwhile, a printout containing medical instructions, in which the type, dosage, frequency, timing, and course of medications and schedules of re-examination were described, was provided and informed to the patients and/or their family members. The possibility of adverse drug reactions (ADRs), such as anaphylactic reactions, bitter taste, darkened stool, and nausea, was also informed. Drugs should be taken sufficiently and regularly each day unless drug withdrawal was indicated by the two physicians. Additionally, the precautions to their lives and diets during therapy were informed as follows: (1) forbid smoking and drinking alcohol; (2) avoid eating spicy, raw, and other irritating foods; (3) wash hands and rinse mouth frequently; (4) advocate individual dining or use serving chopsticks for meals; (5) avoid taking the drugs, particularly tetracycline, with milk (preferably

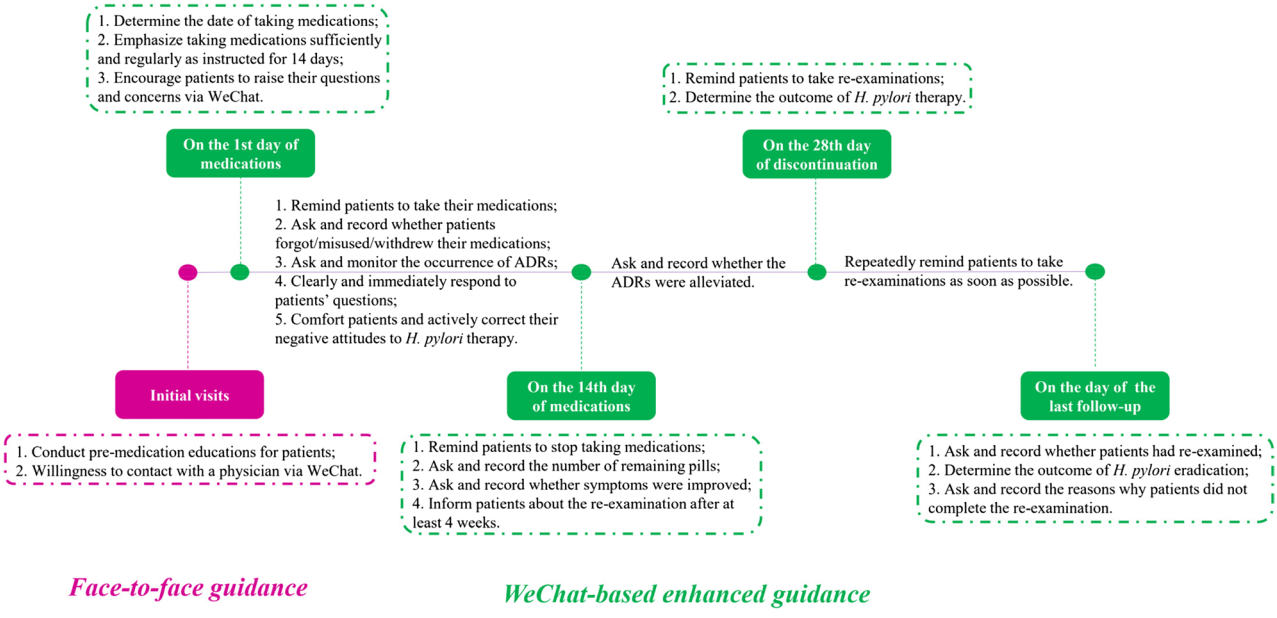


Fig. 1 A flowchart of manual enhanced guidance to patients

at an interval of more than 2 hours); and (6) recommend cohabitants to screen *H. pylori* status [17]. The above contents were understood and accepted by all patients.

Further, the physician (XS) gave the enhanced guidance to each patient via WeChat during a 14-day BQT regimen, and clearly and immediately responded to their concerns about ADRs and forgetfulness/misuse of medications. On the last day of medications, all patients were reminded again to repeat $^{13}C/^{14}C$-UBT as specified.

Data collection

Patients' demographics (i.e., gender and age), height, weight, current status of smoking and/or drinking alcohol, history of peptic ulcer, history of *H. pylori* eradication, family history of gastric cancer, comorbidities, clinical symptoms (i.e., halitosis, belching, acid reflux, heartburn, epigastric pain, bloating, postprandial fullness, early satiation, and/or nausea), dates and approaches of *H. pylori* detection, detection values on $^{13}C/^{14}C$-UBT, and endoscopic and/or pathological reports at their initial visits were collected.

Additionally, detailed information regarding patients' individuals and medications via WeChat during a 14-day BQT regimen was recorded, including place of residence (i.e., rural or urban area), the number of cohabitants, habits of drinking water (i.e., tap water, bottled water, plain boiled water, or tea), style of dining (i.e., collective or individual), smoking, drinking alcohol, education level, dates of starting, discontinuing, and stopping their medications, the number of the remaining pills at the end of their medications, occurrence of ADRs, and remission of clinical symptoms. The aforementioned information was recorded after patients' verbal consents.

Follow up

All patients were followed via WeChat every 2–6 months since the end of a 14-day BQT regimen until repeated $^{13}C/^{14}C$-UBT or July 28, 2022. Telephone visits were only performed in patients who did not response at the last WeChat visits. The dates, results, and detection values of repeated $^{13}C/^{14}C$-UBT and reasons why repeated $^{13}C/^{14}C$-UBT was not performed were recorded.

Evaluation of adherence

In this study, medication adherence [10] was defined as the extent to which patients took the drugs for *H. pylori* eradication as prescribed by physicians, which was evaluated by counting the number of remaining pills at the end of their medications. In detail, good medication adherence would be determined, if a patient actually took ≥ 90% of the pre-specified drugs. Otherwise, poor medication adherence would be determined.

Re-examination adherence was defined as the behavior of repeated $^{13}C/^{14}C$-UBT at least 4 weeks after the end of a 14-day BQT regimen. In detail, good re-examination adherence would be determined, if a patient eventually completed repeated $^{13}C/^{14}C$-UBT. Poor re-examination adherence would be determined, if a patient did not still complete repeated $^{13}C/^{14}C$-UBT at the physician's repeated reminders.

Outcomes

The outcomes of interests included good medication adherence, good re-examination adherence, successful *H. pylori* eradication, and clinical remission. Successful *H. pylori* eradication would be considered, if the result of repeated $^{13}C/^{14}C$-UBT was negative. Clinical remission would be considered, if gastrointestinal symptoms and/or signs at patients' initial visits were improved after *H. pylori* therapy.

Statistical analyses

All statistical analyses were performed using SPSS version 26.0 (IBM Corp, Armonk, NY, USA). Continuous variables were expressed as mean ± standard deviation or median (range), and their difference between groups was compared with T-test or Mann-Whitney U test. Categorical variables were expressed as frequency (percentage), and their difference between groups was compared with Chi-square test or Fisher's exact test. Logistic regression analyses were performed to identify risk factors associated with poor re-examination adherence and *H. pylori* eradication failure. Odds ratios (ORs) with 95% confidence intervals (CIs) were calculated. A two-tailed $P < 0.05$ was considered statistically significant.

Results

Patient characteristics

Overall, 99 of 104 patients with *H. pylori* infection screened were finally enrolled (Fig. 2), of whom 51 (51.5%), 40 (40.4%), and 8 (8.1%) were diagnosed by ^{13}C-UBT, ^{14}C-UBT, and serological test, respectively. The average age was 50.30 ± 13.63 years, and 48 (48.5%) were male. The most common upper gastrointestinal symptom was dyspepsia (55.6%), including epigastric pain, bloating, postprandial fullness, early satiation, and nausea [18], followed by acid reflux and/or heartburn (15.2%), belching (12.1%), and halitosis (6.1%). Additionally, 20 (20.2%) patients presented with abnormal defecation, including unformed stools (n = 7) and disordered defecation (n = 13). Notably, 23 (23.2%) patients were asymptomatic, of whom all were incidentally identified by physical examinations. Sixty-seven (67.7%) patients underwent upper gastrointestinal endoscopy at our department, of whom 18 (26.9%) simultaneously underwent pathological biopsy. Among them, 26.9% (17/67), 50.0% (9/18), and 27.8% (5/18) had peptic ulcer, intestinal metaplasia, and dysplasia, respectively. Eight (8.1%)

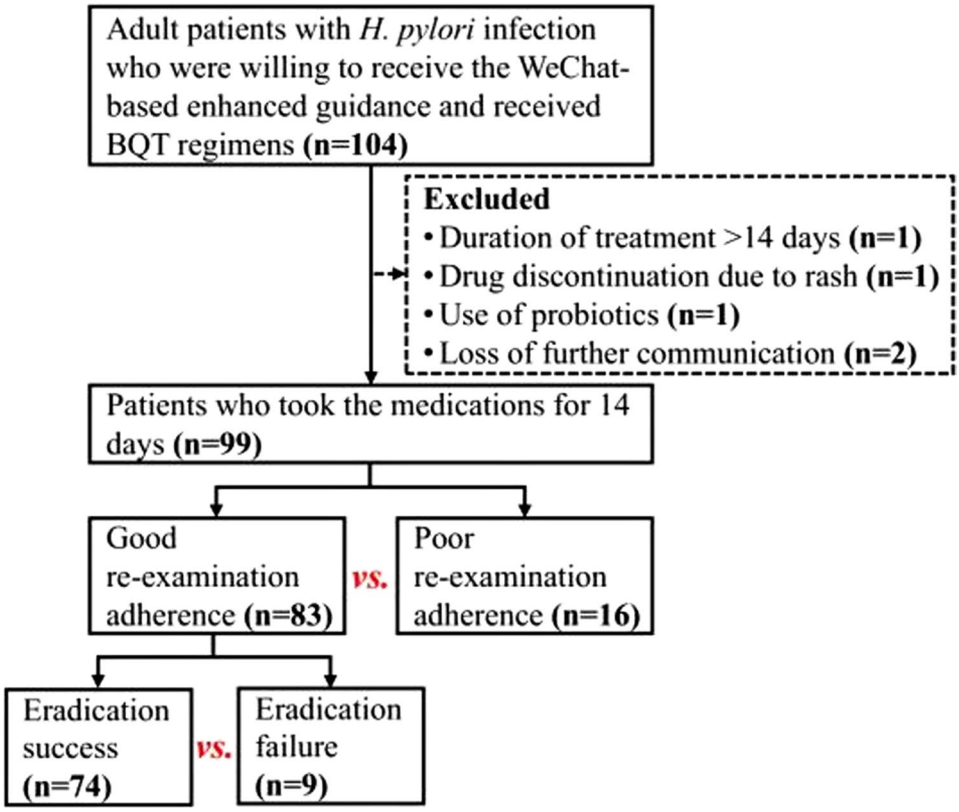


Fig. 2 A flowchart of patient selection

patients had a definite history of *H. pylori* therapy. Clarithromycin-based therapy regimens were prescribed in 93 (93.9%) patients, of whom 91, 1, and 1 received BPCA, BPCM, and BPCT, respectively, and furazolidone-based therapy regimens (BEFT) were prescribed in the remaining 6 (6.1%) patients (Supplementary Table 1).

Adherence

During a 14-day BQT regimen, 12 of 23 smokers stopped smoking and 14 of 15 alcoholic users stopped drinking. Besides, 44 (44.9%) patients selected individual dining and 53 (53.6%) cohabitants were tested for *H. pylori* (Supplementary Table 1).

Medication adherence

Overall, 3 (3.0%) patients had poor medication adherence (Table 1). However, the risk factors for poor medication adherence could not be explored due to a very small number of patients.

Re-examination adherence

Overall, 16 (16.2%) patients did not repeat $^{13}C/^{14}C$-UBT (Fig. 1), of whom 4 were busy and 12 considered it unnecessary (Supplementary Table 2). The remaining 83 patients repeated $^{13}C/^{14}C$-UBT during a median re-examination interval of 48.0 (range: 28.0–559.0) days. Patients with poor re-examination adherence had significantly lower proportions of individual dining (13.3% [2/15] versus 50.6% [42/83], $P=0.008$) and no smoking during treatment (66.7% [10/15] versus 92.8% [77/83], $P=0.012$) than those with good re-examination adherence (Table 1). Univariate logistic regression analysis showed that individual dining (OR = 0.150, 95% CI = 0.032–0.707, $P=0.016$) and no smoking during treatment (OR = 0.156, 95% CI = 0.040–0.606, $P=0.007$) were negatively associated with poor re-examination adherence. Multivariate logistic regression analysis showed that both individual dining (OR = 0.182, 95% CI = 0.038–0.884, $P=0.035$) and no smoking during treatment (OR = 0.210, 95% CI = 0.051–0.865, $P=0.031$) were independently associated with good re-examination adherence (Table 2).

Successful eradication

Eradication rate

Among the 83 patients who repeated $^{13}C/^{14}C$-UBT, the overall rate of successful *H. pylori* eradication was 89.2% (74/83) (Table 1). And the rate of successful *H. pylori* eradication was not significantly different between the WeChat group and the control group (89.2% [74/83] versus 82.6% [438/530], $P=0.137$) (Supplementary Table 3).

Table 1 Difference between good and poor re-examination adherence groups

Variables	Overall		Good re-examination adherence		Poor re-examination adherence		*P*value
	No. Pts	Mean ± SD, Median (range) or Frequency (percentage)	No. Pts	Mean ± SD, Median (range) or Frequency (percentage)	No. Pts	Mean ± SD, Median (range) or Frequency (percentage)	
Male	99	48 (48.5%)	83	37 (44.6%)	16	11 (68.8%)	0.076
Age (years)	99	50.30 ± 13.63	83	51.17 ± 13.74	16	45.81 ± 12.54	0.151
Body mass index (kg/m^2)	99	23.98 ± 3.20	83	24.08 ± 3.14	16	23.43 ± 3.55	0.460
Smoker	98	23 (23.5%)	83	18 (21.7%)	15	5 (33.3%)	0.517
Alcoholic	98	15 (15.3%)	83	12 (14.5%)	15	3 (20.0%)	0.874
History of taking clarithromycin	99	7 (7.1%)	83	5 (6.0%)	16	2 (12.5%)	0.695
History of taking NSAIDs	98	8 (8.2%)	83	7 (8.4%)	15	1 (6.7%)	1.000
Family history of gastric cancer	98	6 (6.1%)	83	6 (7.2%)	15	0 (0%)	0.586
Place of residence(rural area/urban area)	98	13/85(13.3%/86.7%)	83	10/73(12.0%/88.0%)	15	3/12(20.0%/80.0%)	0.673
<3 cohabitants	98	42 (42.9%)	83	36 (43.4%)	15	6 (40.0%)	0.808
Cohabitants with *H. pylori* infection(yes/no)	99	15/38 (15.2%/38.4%)	83	13/34 (15.7%/41.0%)	16	2/4 (12.5%/25.0%)	1.000
Style of dining during treatment(individual/collective)	98	44/54 (44.9%/55.1%)	83	42/41 (50.6%/49.4%)	15	2/13 (13.3%/86.7%)	*>0.008*
Smoking during treatment	98	11 (11.2%)	83	6 (7.2%)	15	5 (33.3%)	**0.012**
Drinking alcohol during treatment	98	1 (1.0%)	83	0 (0%)	15	1 (6.7%)	0.153
Habits of drinking water(raw water/boiled water)	98	34/64(34.7%/65.3%)	83	26/57(31.3%/68.7%)	15	8/7(53.3%/46.7%)	0.099
Education level(high school or below/university or above)	98	58/40(59.2%/40.8%)	83	50/33(60.2%/39.8%)	15	8/7(53.3%/46.7%)	0.616
Gastrointestinal symptoms	99	76 (76.8%)	83	64 (77.1%)	16	12 (75.0%)	1.000
Halitosis	99	6 (6.1%)	83	5 (6.0%)	16	1 (6.3%)	1.000
Belching	99	12 (12.1%)	83	12 (14.5%)	16	0 (0%)	0.229
Acid reflux and/or heartburn	99	15 (15.2%)	83	12 (14.5%)	16	3 (18.8%)	0.954
Dyspeptic symptoms†	99	55 (55.6%)	83	48 (57.8%)	16	7 (43.8%)	0.299
Unformed stools	99	7 (7.1%)	83	4 (4.8%)	16	3 (18.8%)	0.145
Disordered defecation	99	13 (13.1%)	83	11 (13.3%)	16	2 (12.5%)	1.000
Endoscopy	99	67 (67.7%)	83	58 (69.9%)	16	9 (56.3%)	0.286
Esophagitis	67	12 (17.9%)	58	9 (15.5%)	9	3 (33.3%)	0.407
Gastric polyp	67	9 (13.4%)	58	9 (15.5%)	9	0 (0%)	0.456
Peptic ulcer	67	17 (25.4%)	58	15 (25.9%)	9	2 (22.2%)	1.000
Pathological biopsy on endoscopy	67	18 (26.9%)	58	17 (29.3%)	9	1 (11.1%)	0.458
Gastritis	18	13 (72.2%)	17	12 (70.6%)	1	1 (100%)	1.000
Peptic ulcer	18	2 (11.1%)	17	1 (5.9%)	1	1 (100%)	0.111
Intestinal metaplasia and/or dysplasia	18	11 (61.1%)	17	10 (58.8%)	1	1 (100%)	1.000
Initial therapy	99	91 (91.9%)	83	77 (92.8%)	16	14 (87.5%)	0.836
Clarithromycin-/furazolidone-based therapy regimens	99	93/6(93.9%/6.1%)	83	77/6(92.8%/7.2%)	16	16/0(100%/0%)	0.585
Clinical remission	76	68 (89.5%)	64	58 (90.6%)	12	10 (83.3%)	0.808
Halitosis	6	5 (83.3%)	5	4 (80.0%)	1	1 (100%)	1.000
Belching	12	8 (66.7%)	12	8 (66.7%)	0	0 (0%)	—
Acid reflux and/or heartburn	15	13 (86.7%)	12	10 (83.3%)	3	3 (100%)	1.000
Dyspeptic symptoms†	55	49 (89.1%)	48	44 (91.7%)	7	5 (71.4%)	0.163
Unformed stools	7	7 (100%)	4	4 (100%)	3	3 (100%)	—
Disordered defecation	13	13 (100%)	11	11 (100%)	2	2 (100%)	—
Adverse drug reactions	99	56 (56.6%)	83	49 (59.0%)	16	7 (43.8%)	0.259
Poor medication adherence	99	3 (3.0%)	83	3 (3.6%)	16	0 (0%)	1.000

Note. †Dyspeptic symptoms include epigastric pain or burning, bloating, postprandial fullness, early satiation, and/or nausea

Abbreviations: *No. Pts*, Number of patients; *NSAIDs*, Non-steroidal anti-inflammatory drugs

Table 2 Logistic regression analyses of predictors of poor re-examination adherence

Variables	Univariate analysis			Multivariate analysis		
	OR	95%CI	*P*value	OR	95%CI	*P* value
Gender (male vs female)	2.735	0.873–8.572	0.084			
Age (years)	0.971	0.933–1.011	0.153			
Body mass index (kg/m^2)	0.936	0.787–1.113	0.456			
Smoker (yes vs no)	1.806	0.547–5.957	0.332			
Alcoholic (yes vs no)	1.479	0.363–6.030	0.585			
History of taking clarithromycin (yes vs no)	2.229	0.393–12.642	0.366			
History of taking non-steroidal anti-inflammatory drugs (yes vs no)	0.776	0.088–6.802	0.819			
Family history of gastric cancer (yes vs no)	—	—	0.999			
Place of residence (rural area vs urban area)	1.825	0.438–7.606	0.409			
≥3 vs <3 cohabitants	1.149	0.375–3.523	0.808			
Cohabitants with *Helicobacter pylori* infection (yes vs no)	1.308	0.213–8.020	0.772			
Style of dining during treatment (individual vs collective)	0.150	0.032–0.707	*0.016*	0.182	0.038–0.884	*0.035*
Smoking during treatment (no vs yes)	0.156	0.040–0.606	*0.007*	0.210	0.051–0.865	*0.031*
Drinking alcohol during treatment (no vs yes)	—	—	1.000			
Habits of drinking water (raw water vs boiled water)	2.505	0.821–7.644	0.107			
Education level (high school or below vs university or above)	0.754	0.250–2.279	0.617			
Gastrointestinal symptoms (yes vs no)	0.891	0.257–3.084	0.855			
Endoscopy (yes vs no)	0.554	0.186–1.654	0.290			
Esophagitis (yes vs no)	2.722	0.573–12.923	0.208			
Gastric polyp (yes vs no)	—	—	0.999			
Peptic ulcer (yes vs no)	0.819	0.153–4.385	0.816			
Pathological biopsy on endoscopy (yes vs no)	0.301	0.035–2.599	0.275			
Gastritis (yes vs no)	0.394	0.048–3.269	0.389			
Peptic ulcer (yes vs no)	—	—	0.998			
Intestinal metaplasia and/or dysplasia (yes vs no)	—	—	0.999			
Initial therapy vs salvage therapy	0.545	0.100–2.982	0.484			
Clarithromycin- vs furazolidone-based therapy regimens	—	—	0.999			
Clinical remission (yes vs no)	0.517	0.091–2.933	0.457			
Adverse drug reactions (yes vs no)	0.540	0.183–1.590	0.263			
Medication adherence (poor vs good)	—	—	0.999			

Risk factors for H. pylori eradication failure

Patients with *H. pylori* eradication failure had significantly higher proportions of poor medication adherence (22.2% [2/9] versus 1.4% [1/74], $P = 0.030$) and living in rural area (44.4% [4/9] versus 8.1% [6/74], $P = 0.009$) than those with *H. pylori* eradication success (Table 3). Univariate logistic regression analysis showed that poor medication adherence (OR = 20.857, 95% CI = 1.674–259.904.674.904, $P = 0.018$) and living in rural area (OR = 9.067, 95% CI = 1.910–43.033, $P = 0.006$) were significantly associated with *H. pylori* eradication failure. Multivariate logistic regression analysis showed that both poor medication adherence (OR = 20.366, 95% CI = 1.321–314.077, $P = 0.031$) and living in rural area (OR = 8.944, 95% CI = 1.684–47.507, $P = 0.010$) were independently associated with *H. pylori* eradication failure (Table 4).

Clinical remission

Sixty-eight (89.5%) of 76 symptomatic patients achieved clinical remission (Table 1). Notably, the rate of remission of abnormal defecation was 100% (20/20) in patients with good medication adherence.

Discussion

The major findings of our study were that the rates of good medication adherence, good re-examination adherence, successful *H. pylori* eradication, and clinical remission were 97.0%, 83.8%, 89.2%, and 89.5%, respectively, which may suggest that the WeChat-based enhanced guidance has a positive impact on the outcomes of *H. pylori* therapy. This may be related to the two following reasons. First, all our patients received adequate education via face-to-face, which could improve patients' scientific understandings of *H. pylori* and *H. pylori*-related diseases [19], thus strengthening their attentions to the importance of *H. pylori* therapy, recognizing the rationality of prescriptions, reducing excessive concerns about *H. pylori* infection and ADRs, and enhancing the consciousness of self-management [11]. In this setting, patients are more likely to believe the physicians' suggestions, laying a solid foundation for subsequent visits. Furthermore, a

Table 3 Difference between eradication success and failure groups

Variables	Overall		Eradication success		Eradication failure		*P*value
	No.Pts	Mean ± SD, Median (range) or Frequency (percentage)	No.Pts	Mean ± SD, Median (range) or Frequency (percentage)	No.Pts	Mean ± SD, Median (range) or Frequency (percentage)	
Male	83	37 (44.6%)	74	33 (44.6%)	9	4 (44.4%)	1.000
Age (years)	83	51.17 ± 13.74	74	51.62 ± 14.19	9	47.44 ± 8.89	0.239
Body mass index (kg/m^2)	83	24.08 ± 3.14	74	24.11 ± 3.20	9	23.85 ± 2.73	0.813
Smoker	83	18 (21.7%)	74	17 (23.0%)	9	1 (11.1%)	0.699
Alcoholic	83	12 (14.5%)	74	11 (14.9%)	9	1 (11.1%)	1.000
History of taking clarithromycin	83	5 (6.0%)	74	4 (5.4%)	9	1 (11.1%)	0.445
History of taking NSAIDs	83	7 (8.4%)	74	6 (8.1%)	9	1 (11.1%)	0.567
Family history of gastric cancer	83	6 (7.2%)	74	6 (8.1%)	9	0 (0%)	1.000
Place of residence (rural area/urban area)	83	10/73(12.0%/88.0%)	74	6/68(8.1%/91.9%)	9	4/5(44.4%/55.6%)	*0.009*
<3 cohabitants	83	36 (43.4%)	74	32 (43.2%)	9	4 (44.4%)	1.000
Cohabitants with *H. pylori* infection (yes/no)	83	13/34 (15.7%/41.0%)	74	10/32 (13.5%/43.2%)	9	3/2 (33.3%/22.2%)	0.237
Style of dining during treatment (individual/collective)	83	42/41 (50.6%/49.4%)	74	39/35 (52.7%/47.3%)	9	3/6 (33.3%/66.7%)	0.457
Smoking during treatment	83	6 (7.2%)	74	5 (6.8%)	9	1 (11.1%)	0.509
Drinking alcohol during treatment	83	0 (0%)	74	0 (0%)	9	9 (0%)	—
Habits of drinking water (raw water/boiled water)	83	26/57(31.3%/68.7%)	74	23/51(31.1%/68.9%)	9	3/6(33.3%/66.7%)	1.000
Habits of drinking tea	83	6 (7.2%)	74	5 (6.8%)	9	1 (11.1%)	0.509
Education level (high school or below/university or above)	83	50/33(60.2%/39.8%)	74	44/30(59.5%/40.5%)	9	6/3(66.7%/33.3%)	0.955
Gastrointestinal symptoms	83	64 (77.1%)	74	57 (77.0%)	9	7 (77.8%)	1.000
Halitosis	83	5 (6.0%)	74	5 (6.8%)	9	0 (0%)	1.000
Belching	83	12 (14.5%)	74	11 (14.9%)	9	1 (11.1%)	1.000
Acid reflux and/or heartburn	83	12 (14.5%)	74	10 (13.5%)	9	2 (22.2%)	0.842
Dyspeptic symptoms†	83	48 (57.8%)	74	43 (58.1%)	9	5 (55.6%)	1.000
Unformed stools	83	4 (4.8%)	74	4 (5.4%)	9	0 (0%)	1.000
Disordered defecation	83	11 (13.3%)	74	11 (14.9%)	9	0 (0%)	0.471
Endoscopy	83	58 (69.9%)	74	51 (68.9%)	9	7 (77.8%)	0.871
Esophagitis	58	9 (15.5%)	51	7 (13.7%)	7	2 (28.6%)	0.645
Gastric polyp	58	9 (15.5%)	51	9 (17.6%)	7	0 (0%)	0.514
Peptic ulcer	58	15 (25.9%)	51	14 (27.5%)	7	1 (14.3%)	0.775
Pathological biopsy on endoscopy	58	17 (29.3%)	51	17 (33.3%)	7	0 (0%)	0.169
Gastritis	17	12 (70.6%)	17	12 (70.6%)	0	0 (0%)	—
Peptic ulcer	17	1 (5.9%)	17	1 (5.9%)	0	0 (0%)	—
Intestinal metaplasia and/or dysplasia	17	10 (58.8%)	17	10 (58.8%)	0	0 (0%)	—
Initial therapy	83	77 (92.8%)	74	69 (93.2%)	9	8 (88.9%)	0.509
Clarithromycin-/furazolidone-based regimens	83	77/6(92.8%/7.2%)	74	68/6(91.9%/8.1%)	9	9/0(100.0%/0%)	1.000
Clinical remission	64	58 (90.6%)	57	53 (93.0%)	7	5 (71.4%)	0.125
Halitosis	5	4 (80.0%)	5	4 (80.0%)	0	0 (0%)	—
Belching	12	8 (66.7%)	11	8 (72.7%)	1	0 (0%)	0.333
Acid reflux and/or heartburn	12	10 (83.3%)	10	9 (90.0%)	2	1 (50.0%)	0.318
Dyspeptic symptoms†	48	44 (91.7%)	43	40 (93.0%)	5	4 (80.0%)	0.366
Unformed stools	4	4 (100.0%)	4	4 (100.0%)	0	0 (0%)	—
Disordered defecation	11	11 (100.0%)	11	11 (100.0%)	0	0 (0%)	—
Adverse drug reactions	83	49 (59.0%)	74	43 (58.1%)	9	6 (66.7%)	0.893
Poor medication adherence	83	3 (3.6%)	74	1 (1.4%)	9	2 (22.2%)	*0.030*

Note. †Dyspeptic symptoms include epigastric pain or burning, bloating, postprandial fullness, early satiation, and/or nausea

Abbreviations: *No. Pts*, Number of patients; *NSAIDs*, Non-steroidal anti-inflammatory drugs; *H. pylori*, Helicobacter pylori

Table 4 Logistic regression analyses of predictors of eradication failure

Variables	Univariate analysis			Multivariate analysis		
	OR	95%CI	*P* value	OR	95%CI	*P* value
Gender (male vs female)	0.994	0.247–4.000	0.993			
Age (years)	0.978	0.929–1.029	0.390			
Body mass index (kg/m^2)	0.973	0.776–1.219	0.810			
Smoker (yes vs no)	0.419	0.049–3.592	0.428			
Alcoholic (yes vs no)	0.716	0.081–6.303	0.763			
History of taking clarithromycin (yes vs no)	2.187	0.217–22.042	0.507			
History of taking non-steroidal anti-inflammatory drugs (yes vs no)	1.417	0.151–13.310	0.761			
Family history of gastric cancer (yes vs no)	—	—	0.999			
Place of residence (rural area vs urban area)	9.067	1.910–43.033	*0.006*	8.944	1.684–47.507	*0.010*
≥3 vs <3 cohabitants	0.952	0.237–3.835	0.945			
Cohabitants with *Helicobacter pylori* infection (yes vs no)	4.800	0.700–32.903	0.110			
Style of dining during treatment (individual vs collective)	0.449	0.104–1.930	0.282			
Smoking during treatment (no vs yes)	0.580	0.060–5.602	0.638			
Drinking alcohol during treatment (no vs yes)	—	—	—			
Habits of drinking water (raw water vs boiled water)	1.109	0.255–4.826	0.891			
Habits of drinking tea (yes vs no)	1.725	0.179–16.670	0.638			
Education level (high school or below vs university or above)	1.364	0.316–5.881	0.677			
Gastrointestinal symptoms (yes vs no)	1.044	0.198–5.502	0.960			
Endoscopy (yes vs no)	1.578	0.304–8.192	0.587			
Esophagitis (yes vs no)	2.514	0.406–15.572	0.322			
Gastric polyp (yes vs no)	—	—	0.999			
Peptic ulcer (yes vs no)	0.440	0.049–3.993	0.466			
Pathological biopsy on endoscopy (yes vs no)	—	—	0.998			
Gastritis (yes vs no)	—	—	0.999			
Peptic ulcer (yes vs no)	—	—	1.000			
Intestinal metaplasia and/or dysplasia (yes vs no)	—	—	0.999			
Initial therapy vs salvage therapy	0.580	0.060–5.602	0.638			
Clarithromycin- vs furazolidone-based therapy regimens	—	—	0.999			
Clinical remission (yes vs no)	0.189	0.027–1.299	0.090			
Adverse drug reactions (yes vs no)	1.442	0.335–6.214	0.623			
Medication adherence (poor vs good)	20.857	1.674–259.904	*0.018*	20.366	1.321–314.077	*0.031*

recent meta-analysis suggested [11] that the enhanced patient education group had significantly better adherence than the control group (RR = 1.48, 95%CI = 1.14–1.93, P = 0.003). Second, WeChat provides a convenient and costless platform for real-time interaction between people [15, 16]. Besides, enhanced patient education could significantly improve the rates of *H. pylori* eradication in the WeChat subgroup (RR = 1.24, 95%CI = 1.03–1.50, P = 0.02), but not in the telephone calls subgroup (RR = 1.12, 95%CI = 0.91–1.38, P = 0.29) or short message subgroup (RR = 1.12, 95%CI = 0.99–1.27, P = 0.08) [11]. In our study, WeChat is served as a reminder of taking medicines regularly and repeating $^{13}C/^{14}C$-UBT. In detail, patients could get timely responses and professional guidance from the physician via WeChat for any problem that they encountered during *H. pylori* therapy, reducing the use of incorrect drugs/dosages and the behavior of drug withdrawal without permissions, thereby contributing to successful *H. pylori* eradication.

Our study found a higher rate of successful *H. pylori* eradication in the WeChat group than in the control group without WeChat management, with an increase of approximately 7%. This may be mainly due to the three following reasons. First, all our patients were given enhanced guidance via WeChat, greatly increasing the rationality and accuracy of drug use, thereby improving successful *H. pylori* eradication. By contrast, patients only received traditional instructions at their initial visits in our previous study. Second, therapeutic regimen was prescribed based on the patient's history of *H. pylori* therapy and/or medication use, which decreased the risk of eradication failure due to the use of antibiotics with high resistance to *H. pylori*, mainly clarithromycin, metronidazole, and levofloxacin [20]. By contrast, neither patient's history of *H. pylori* therapy nor medication use was known in our previous study. Third, all our treatment-experienced patients were treated at least 3 months after the end of the last medications, which is beneficial for the transition of coccoid form, a negative state of *H.*

pylori, to rod-shaped form, an active state of *H. pylori*, promoting the redistribution of *H. pylori* in the stomach, and increasing the exposure and sensitivity of the drugs, thus exerting an effective bactericidal role [21, 22]. By contrast, the interval between two consecutive *H. pylori* therapy was unknown in our previous study.

It remains controversial whether the application of WeChat platform is effective for *H. pylori* therapy. Four studies can be searched in the PubMed database [19, 23–25]. Luo et al.[19] and Ma et al.[23] affirmed the value of WeChat management on *H. pylori* therapy compared to traditional instructions. Specifically, Luo et al.[19] found that the intervention group had significantly better medication adherence (94.6% versus 54.1%, $P < 0.0001$) and higher rate of successful *H. pylori* eradication (87.4% versus 63.1%, $P < 0.0001$) than the control group. Similarly, Ma et al.[23] found that the intervention group had significantly better medication adherence ($P < 0.05$) and higher rates of re-examination (93.8% versus 77.6%, $P < 0.05$) and successful *H. pylori* eradication (90.0% versus 78.6%, $P < 0.05$) than the control group. Furthermore, Sun et al.[24] found that the application of WeChat platform only significantly improved medication adherence (ITT analysis: 85.52% versus 70.48%, $P = 0.028$; PP analysis: 92.39% versus 81.90%, $P = 0.030$), but not *H. pylori* eradication (ITT analysis: 76.70% versus 70.73%, $P = 0.312$; PP analysis: 85.87% versus 82.86%, $P = 0.562$) compared to one-time verbal education, while Lin et al.[25] found that the application of WeChat platform did not yield significantly better results compared to conventional patient education. However, history of antibiotic use [24], number of cohabitants [19, 24], history of *H. pylori* eradication [23], and education level [25] were not collected in the four studies. In this setting, the difference of patients' characteristics between the intervention group and the control group might affect the results. By contrast, our study recorded more comprehensive data on patients' characteristics at baseline, treatment, and clinical remission. Moreover, our study showed that the rates of patients' medication adherence and successful *H. pylori* eradication under the WeChat-based enhanced guidance were similar and even higher than those of the intervention groups in the four studies.

Another finding of our study was that not all patients repeated $^{13}C/^{14}C$-UBT, probably due to their cognitive deviations [19] and physicians' motivations [9]. In general, *H. pylori* infection and its-related diseases can be cured/alleviated by correctly taking medications. However, it should be highlighted that clinical symptom remission is not equal to *H. pylori* eradication. According to the Maastricht VI/Florence consensus [3] and the sixth Chinese consensus [26], a repeated $^{13}C/^{14}C$-UBT 4 weeks after the end of medications is required to determine the outcomes of *H. pylori* therapy. Otherwise, *H. pylori* eradication failure might increase the risk of disease progression or recurrence, especially in patients with peptic ulcer, atrophy, intestinal metaplasia, dysplasia, and a family history of gastric cancer. Thus, the importance of repeated $^{13}C/^{14}C$-UBT should be sufficiently recognized and needs to be further emphasized in clinical practice.

Our study also demonstrated that abnormal defecation was improved in patients with good medication adherence. This may be because *H. pylori* therapy is beneficial to repair damaged gastric mucosa, thereby improving dyspepsia [27] and abnormal defecation [28]. On the other hand, *H. pylori* infection can cause gastrointestinal dysfunction, such as irritable bowel syndrome (IBS) [29]. Notably, IBS symptoms can be alleviated by *H. pylori* therapy [30]. However, it should be acknowledged that only a small number of patients with abnormal defecation are included in our study. Therefore, further studies are needed to confirm this finding.

Our study has several features. First, all patients were treated by the same therapeutic strategy and followed via WeChat by an investigator, which could minimize the heterogeneity caused by different investigators and therapeutic strategies. Second, WeChat provides a novel approach for the management of *H. pylori*-related diseases, greatly improving the efficiency of patient-physician communication. Third, one-by-one online communication during *H. pylori* therapy was employed for individualized guidance in our all patients [26].

There were also several limitations. First, WeChat-based enhanced guidance is not applicable to patients without smartphones, which may produce a selection bias at the time of patient enrollment. Second, three different approaches were employed to identify *H. pylori* infection. We acknowledge that there may be differences in diagnostic accuracy between approaches. Third, the number of patients was small in this study, increasing the possibility of type II errors in statistics. Fourth, the number of the remaining pills could not be verified in some patients, which might lead to a few deviations in our judgement of medication adherence. Fifth, the re-examination interval was relatively long in some patients, which might potentially increase the risk of *H. pylori* reinfection/recurrence, contributing to the probability of positive results at repeated $^{13}C/^{14}C$-UBT, thereby underestimating the rate of *H. pylori* successful eradication. Nevertheless, the annual *H. pylori* reinfection rates were low with 3.1% globally[31] and 1.5% in China [32].

Conclusion

In conclusion, medication adherence is good, and the rate of successful *H. pylori* eradication is high under the WeChat-based enhanced guidance, but re-examination adherence should be further improved. Given the

advantages of WeChat management, it could be considered as an auxiliary tool to guide *H. pylori* eradication therapy.

Abbreviations

H. pylori	*Helicobacter pylori*
UBT	Urea breath test
BQT	Bismuth-containing quadruple therapy
CBP	Colloidal bismuth pectin
PPI	Proton pump inhibitor
ADRs	Adverse drug reactions
IBS	Irritable bowel syndrome

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Trial registration statement
This study is a retrospective observational study based on the database established by our study team. Therefore, the trial registration is not required.

Authors' contributions
Author contributions: Conceptualization: X.Q.; Formal analysis: X.S. and X.Q.; Data curation: X.S., W.X., and X.Q.; Data accuracy: X.S., Z.L., and W.X.; Writing-original draft: X.S. and X.Q.; Writing-review and editing: X.S., Z.L., Z.T., C.Y., W.X., H.L., and X.Q.;Supervision: X.Q.

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Data availability
The datasets used and/or analyzed during the current study are available fromthe corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was performed according to the Declaration of Helsinki and was approved by the Medical Ethical Committee of the General Hospital of Northern Theater Command (approval number: Y (2023) 056). The requirement of informed written consent for this study was waived due to its retrospective nature.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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